

MATHEMATICAL POSTULATES OF QUANTUM MECHANICS

Unit 3: The Postulates of Quantum Mechanics

The four rules that turn the vectors and matrices of Unit 2 into a working theory of qubits, with the notation physicists use to write them.

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COURSE

25AID432 Mathematical Postulates of
Quantum Mechanics

UNIT

3 of 3

SESSIONS

10 to 15

PREREQUISITE

Units 1 and 2, including eigenvalues and the
tensor product

READING TIME

About two and a half hours

OFFERING

Odd Semester 2026-27

Before you start

Units 1 and 2 were mathematics. This unit is physics, but you will find very little new mathematics in it. Almost everything here is Unit 2 written in a new notation, plus four rules that say what the vectors and matrices *mean*.

Those four rules are called **postulates**. A postulate is a rule we accept without deriving it from anything more basic. We accept it because every prediction made with it has matched experiment, for about a hundred years and across millions of measurements. Nobody proved that a measurement probability is the squared modulus of an inner product. Max Born proposed it in 1926, people tested it, and it has never failed. So when you ask “why is it squared?”, the honest answer in this course is “because that is what nature does”. What you *can* check is that the rules hang together: that probabilities come out between 0 and 1, that they add to 1, and that the rules do not contradict each other. We will check exactly those things.

The mathematics you need is all from Unit 2. Make sure you are comfortable with the conjugate transpose (dagger), the inner product with the **first** argument conjugated, norms and normalisation, orthonormal bases, projection, Hermitian and unitary matrices, eigenvalues and eigenvectors, and the Kronecker (tensor) product. If any of those feels shaky, reread that section of Unit 2 first.

BY THE END OF THIS UNIT YOU WILL BE ABLE TO

- State the four postulates of quantum mechanics and write each as a formula.
- Translate between the vector notation of Unit 2 and bra-ket notation, and compute brackets, outer products and projectors.
- Write a qubit state, normalise it, and decide whether two kets describe the same physical state (global phase) or different ones (relative phase).
- Use the Born rule to compute measurement probabilities in any orthonormal basis, and give the state after the measurement.
- Apply the gates X, Y, Z, H and S and sequences of them to a qubit, in the correct order.
- Build two-qubit states with the tensor product, apply CNOT, create a Bell state, and prove that it is entangled.
- Treat a Hermitian matrix as an observable: find its possible outcomes, their probabilities, and its expectation value.
- Check each of these calculations in a few lines of NumPy.

KEY TERMS IN THIS UNIT

postulate · ket · bra · bracket · outer product · projector · completeness relation · qubit · computational basis · superposition · normalisation · amplitude · global phase · relative phase · Bloch sphere · measurement · Born rule · collapse · measurement basis · unitary evolution · quantum gate

Pauli gates · Hadamard gate · phase gate · quantum circuit · interference · composite system · tensor product · product state · CNOT · entanglement · Bell state · observable · eigenvalue · expectation value · variance

The postulates at a glance

Here is the whole unit on one page. Every later section expands one row of this table.

NO.	NAME	WHAT IT SAYS	IN SYMBOLS
1	State	A system is described by a unit vector in a complex vector space. Kets that differ only by a global phase describe the same physical state.	$ \psi\rangle$ with $\langle\psi \psi\rangle = 1$; $e^{i\gamma} \psi\rangle \equiv \psi\rangle$
2	Evolution	A closed system changes by a unitary matrix.	$ \psi\rangle \rightarrow U \psi\rangle$, $U^\dagger U = I$
3	Measurement	Measuring in an orthonormal basis gives outcome k with probability equal to the squared modulus of an inner product; the state afterwards is the basis vector found.	$p(k) = \langle b_k \psi\rangle ^2$, then $ \psi\rangle \rightarrow b_k\rangle$
4	Composite systems	The state space of a joint system is the tensor product of the state spaces of its parts.	$\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$

The last section of the unit adds **observables**. An observable is Postulate 3 said again in the language of Hermitian matrices: a physical quantity is a Hermitian matrix A , the possible results are its eigenvalues, and the average result is $\langle\psi|A|\psi\rangle$.

NOTE · ABOUT THE NUMBERING

Textbooks list the postulates in different orders. Some have five or six, because they split measurement into two rules or give observables a number of their own. The content is the same everywhere. In this course, in the assignment and in the exams, the numbers are the ones in the table above.

The six sessions of this unit take the material in the order of this booklet.

SESSION	TOPIC IN THIS BOOKLET
10	The postulates at a glance; bra-ket notation
11	The single quantum state and the qubit; superposition; global and relative phase
12	Quantum measurement: the Born rule, collapse, measuring in another basis
13	Quantum operations: unitary evolution, the gates X, Y, Z, H, S, sequences of gates
14	Multiple quantum states: two qubits, product states, CNOT, entanglement
15	Observables and expectation values; putting it together

Bra-ket notation

Physicists write vectors in a notation invented by Paul Dirac, called **bra-ket notation**. It looks strange for about a week and then becomes the easiest way to write anything in quantum mechanics. Nothing about it is new mathematics.

Kets and bras

DEFINITION · KET AND BRA

A **ket** $|\psi\rangle$ is a column vector. The symbol inside, here ψ , is only a name. The vector $\vec{v}$ from Unit 2 is $|v\rangle$ here; nothing new except the notation.

A **bra** $\langle\psi|$ is the conjugate transpose (dagger) of the ket: $\langle\psi| = |\psi\rangle^\dagger$. It is a row vector whose entries are the complex conjugates of the ket's entries.

For example,

$$|\psi\rangle = \begin{pmatrix} \frac{3}{5} \\ \frac{4i}{5} \end{pmatrix} \implies \langle\psi| = \left(\frac{3}{5} \quad -\frac{4i}{5} \right).$$

The ket is a column, the bra is a row, and the i became $-i$ on the way.

The bracket is the inner product

Put a bra and a ket together and you get a **bracket**, $\langle\phi|\psi\rangle$. It is a row vector times a column vector, which is a single complex number:

$$\langle\phi|\psi\rangle = |\phi\rangle^\dagger |\psi\rangle = \sum_k \phi_k^* \psi_k.$$

This is exactly the inner product $\langle\vec{\phi}, \vec{\psi}\rangle$ of Unit 2, with the first vector conjugated. The notation now does the conjugating for you: whatever sits on the bra side is conjugated, because a bra is a dagger.

Two facts from Unit 2 carry over directly:

$$\langle\phi|\psi\rangle^* = \langle\psi|\phi\rangle, \quad \langle\psi|\psi\rangle = |\psi|^2 \geq 0.$$

Swapping the two sides of a bracket conjugates it. And a ket is **normalised** when $\langle\psi|\psi\rangle = 1$.

PITFALL · WHICH SIDE GETS CONJUGATED

In this course the **bra** side is conjugated. Many mathematics books conjugate the second argument instead, and with complex vectors the two conventions give answers that are conjugates of each other. When you meet a formula from another source, check which convention it uses before you trust a sign.

WORKED EXAMPLE · 1. TWO BRACKETS

Take $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $|+i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$.

(a) $\langle + | - \rangle = \frac{1}{2} (1 \cdot 1 + 1 \cdot (-1)) = 0$. The two kets are orthogonal.

(b) For $\langle +i | + \rangle$ the bra is $\frac{1}{\sqrt{2}} (1, -i)$, so $\langle +i | + \rangle = \frac{1}{2} (1 - i)$.

(c) Swapping sides: $\langle + | +i \rangle = \frac{1}{2} (1 + i)$, which is the conjugate of (b), as the rule says.

The outer product and projectors

A bra times a ket is a number. A ket times a bra, $|\psi\rangle\langle\phi|$, is a column times a row, which is a **matrix**. It is called the **outer product**. For two-entry vectors,

$$|\psi\rangle\langle\phi| = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} (\phi_1^* \quad \phi_2^*) = \begin{pmatrix} \psi_1\phi_1^* & \psi_1\phi_2^* \\ \psi_2\phi_1^* & \psi_2\phi_2^* \end{pmatrix}.$$

The notation tells you how the matrix acts. Apply $|\psi\rangle\langle\phi|$ to a ket $|\chi\rangle$ and the bra meets the ket to make a number:

$$(|\psi\rangle\langle\phi|) |\chi\rangle = |\psi\rangle \langle\phi|\chi\rangle = \langle\phi|\chi\rangle |\psi\rangle.$$

The special case $|b\rangle\langle b|$, with $|b\rangle$ a unit vector, is the **projector** onto $|b\rangle$. It takes any ket to its component along $|b\rangle$: $|b\rangle\langle b|\psi\rangle = \langle b|\psi\rangle |b\rangle$. This is the projection of Unit 2 (session 7), now written in one line.

WORKED EXAMPLE · 2. OUTER PRODUCTS AS MATRICES

(a) $|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (0 \quad 1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. It sends $|1\rangle$ to $|0\rangle$ and sends $|0\rangle$ to the zero vector.

(b) $|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1 \quad 1) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

(c) Project $|0\rangle$ onto $|+\rangle$. Using the matrix: $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$. Using the notation: $\langle + | 0 \rangle |+\rangle = \frac{1}{\sqrt{2}} |+\rangle$, which is the same vector. The second route needs no matrix at all.

The completeness relation

If $\{|b_1\rangle, |b_2\rangle, \dots, |b_n\rangle\}$ is an orthonormal basis, the projectors onto the basis vectors add up to the identity:

$$\sum_{k=1}^n |b_k\rangle\langle b_k| = I.$$

This is the **completeness relation**. Apply both sides to any $|\psi\rangle$:

$$|\psi\rangle = \sum_k |b_k\rangle\langle b_k|\psi\rangle = \sum_k \langle b_k|\psi\rangle |b_k\rangle.$$

That is the statement from Unit 2 that the components of a vector in an orthonormal basis are its inner products with the basis vectors. You will use this every time you rewrite a state in another basis.

WORKED EXAMPLE · 3. COMPLETENESS IN THE $\pm$ BASIS

From Worked example 2, $|+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. In the same way $|-\rangle\langle-| = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Adding,

$$|+\rangle\langle+| + |-\rangle\langle-| = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

So $\{|+\rangle, |-\rangle\}$ is a complete orthonormal basis of $\mathbb{C}^2$, and any qubit can be written $|\psi\rangle = \langle+|\psi\rangle |+\rangle + \langle-|\psi\rangle |-\rangle$.

From Unit 2 to Unit 3: a translation table

IDEA	UNIT 2 NOTATION	UNIT 3 NOTATION
Column vector	$\vec{v}$	$ v\rangle$
Conjugate transpose of a vector	$\vec{v}^\dagger$	$\langle v $
Inner product	$\langle \vec{u}, \vec{v} \rangle = \vec{u}^\dagger \vec{v}$	$\langle u v\rangle$
Norm squared	$ \vec{v} ^2$	$\langle v v\rangle$
Matrix acting on a vector	$A\vec{v}$	$A v\rangle$
Component along a unit vector	$\langle \vec{b}, \vec{v} \rangle$	$\langle b v\rangle$
Projection onto a unit vector	$\langle \vec{b}, \vec{v} \rangle \vec{b}$	$ b\rangle\langle b v\rangle$
Sandwich of a matrix	$\vec{u}^\dagger A \vec{v}$	$\langle u A v\rangle$
Kronecker product	$\vec{u} \otimes \vec{v}$	$ u\rangle \otimes v\rangle = u\rangle v\rangle = uv\rangle$

NOTE · CHECK IT IN NUMPY

A ket is a one-dimensional NumPy array. `np.vdot(a, b)` conjugates its **first** argument, so it computes $\langle a|b\rangle$ in exactly our convention. In Python the imaginary unit is written `1j`.

```
import numpy as np
s = 1 / np.sqrt(2)
plus = s * np.array([1, 1])
plusi = s * np.array([1, 1j])
print(np.vdot(plusi, plus))           # <+i|+> = (0.5-0.5j)
print(np.outer(plus, plus.conj()))    # |+><+|, a 2x2 matrix
minus = s * np.array([1, -1])
print(np.outer(plus, plus.conj()) + np.outer(minus, minus.conj())) # I
```

The single quantum state and the qubit

Postulate 1 for one qubit

DEFINITION · QUBIT

A **qubit** is a quantum system whose state space is $\mathbb{C}^2$. Its state is a unit vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The kets $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form the **computational basis**. The complex numbers α and β are called **amplitudes**.

A physical qubit can be the spin of an electron, the polarisation of a photon, or two energy levels of an ion or a superconducting circuit. The mathematics is the same for all of them, which is why this course never needs to say which.

When both amplitudes are non-zero the state is called a **superposition** of $|0\rangle$ and $|1\rangle$. The word describes a vector that is a combination of two basis vectors, $\alpha|0\rangle + \beta|1\rangle$, and nothing more mysterious.

The condition $|\alpha|^2 + |\beta|^2 = 1$ is **normalisation**. It says the vector has length 1, and in the next section you will see that it also says the measurement probabilities add to 1. If someone hands you a vector that is not of length 1, divide it by its norm.

WORKED EXAMPLE · 4. NORMALISING A STATE

Normalise $|0\rangle + 2i|1\rangle$.

Norm squared: $|1|^2 + |2i|^2 = 1 + 4 = 5$. So the norm is $\sqrt{5}$ and the normalised state is

$$|\psi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2i|1\rangle).$$

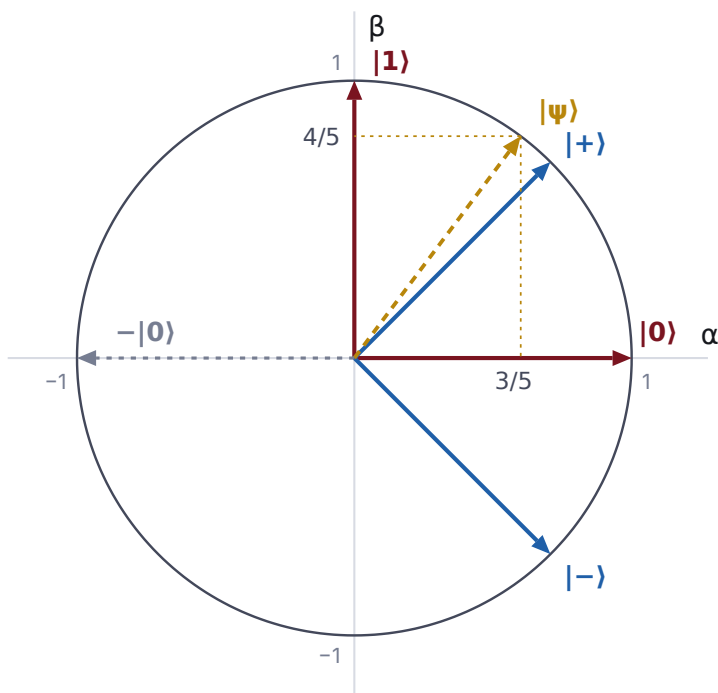
Check: $\left|\frac{1}{\sqrt{5}}\right|^2 + \left|\frac{2i}{\sqrt{5}}\right|^2 = \frac{1}{5} + \frac{4}{5} = 1$. Remember that $|2i|^2 = (2i)^*(2i) = (-2i)(2i) = 4$. Writing $(2i)^2 = -4$ instead is the most common slip in this whole unit.

Six states worth knowing by name

NAME	AS A COMBINATION	AS A COLUMN	WHERE IT COMES UP
$ 0\rangle$	$ 0\rangle$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	computational basis
$ 1\rangle$	$ 1\rangle$	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	computational basis
$ +\rangle$	$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$H 0\rangle$
$ -\rangle$	$\frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$	$H 1\rangle$
$ +i\rangle$	$\frac{1}{\sqrt{2}}(0\rangle + i 1\rangle)$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$	$S +\rangle$
$ -i\rangle$	$\frac{1}{\sqrt{2}}(0\rangle - i 1\rangle)$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$	$S -\rangle$

They come in three orthonormal pairs: $\{|0\rangle, |1\rangle\}$, $\{|+\rangle, |-\rangle\}$ and $\{|+i\rangle, |-i\rangle\}$. Each pair is a basis of $\mathbb{C}^2$, and each gives a different way to measure a qubit.

When the amplitudes are real, a qubit state is a point on the unit circle, since $\alpha^2 + \beta^2 = 1$. Figure 3.1 draws the real states from the table. Keep in mind that the picture shows only part of the story, because in general the amplitudes are complex.



Real amplitudes only

A state $\alpha|0\rangle + \beta|1\rangle$ with real α, β is a point on the unit circle, because $\alpha^2 + \beta^2 = 1$.

$|0\rangle$ and $|1\rangle$ are at right angles: $\langle 0|1\rangle = 0$. So are $|+\rangle$ and $|-\rangle$.

$$|\psi\rangle = (3/5)|0\rangle + (4/5)|1\rangle$$

$-|0\rangle$ differs from $|0\rangle$ only by the global phase -1 , so it is the same physical state.

Figure 3.1 Qubit states with real amplitudes. The horizontal axis is the amplitude of $|0\rangle$ and the vertical axis the amplitude of $|1\rangle$. Orthogonal kets meet at a right angle. The dashed arrow $-|0\rangle$ points the opposite way from $|0\rangle$ but is the same physical state, because the two differ only by the global phase -1 .

Global phase: the part of a ket that does not matter

Here is the second half of Postulate 1. Multiply a ket by a complex number of modulus 1, $e^{i\gamma}$. The result $e^{i\gamma}|\psi\rangle$ is a different vector, but it describes **the same physical state**. The factor $e^{i\gamma}$ is called a **global phase**, because it multiplies every amplitude at once.

Why accept this? Because no experiment can tell the two apart, and you can prove that from the measurement rule you will meet in the next section. Every measurement probability has the form $|\langle b|\psi\rangle|^2$. Replace $|\psi\rangle$ by $e^{i\gamma}|\psi\rangle$:

$$|\langle b|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}|^2 |\langle b|\psi\rangle|^2 = 1 \cdot |\langle b|\psi\rangle|^2.$$

The factor drops out of every probability, for every basis. A gate does not help either: $U(e^{i\gamma}|\psi\rangle) = e^{i\gamma}(U|\psi\rangle)$, so the global phase rides along unchanged and drops out of the next measurement. A difference that never shows up in any experiment is not a physical difference.

Relative phase: the part that does

Now compare $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The amplitudes have the same moduli. The only difference is the factor $-1 = e^{i\pi}$ on the $|1\rangle$ amplitude alone. A phase that sits between the amplitudes like this is a **relative phase**, and it is physical. $|+\rangle$ and $|-\rangle$ are orthogonal, so they are as different as two qubit states can be. In the measurement section you will see an experiment that separates them every time.

The rule: you may pull a common factor out of **all** the amplitudes and throw it away. You may not change the phase of **one** amplitude and keep the other.

WORKED EXAMPLE · 5. SAME STATE OR DIFFERENT?

(a) $\frac{1}{\sqrt{2}}(i|0\rangle + i|1\rangle)$ and $|+\rangle$. The first is $i \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = e^{i\pi/2}|+\rangle$. Only a global phase separates them, so they are the **same** physical state.

(b) $-|1\rangle$ and $|1\rangle$. Global phase -1 : the **same** state.

(c) $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $|+\rangle$. To turn one into the other you would multiply the $|1\rangle$ amplitude alone by i . No single factor multiplies both amplitudes to do it, so this is a relative phase and they are **different** states. (In Worked example 9 a measurement tells them apart.)

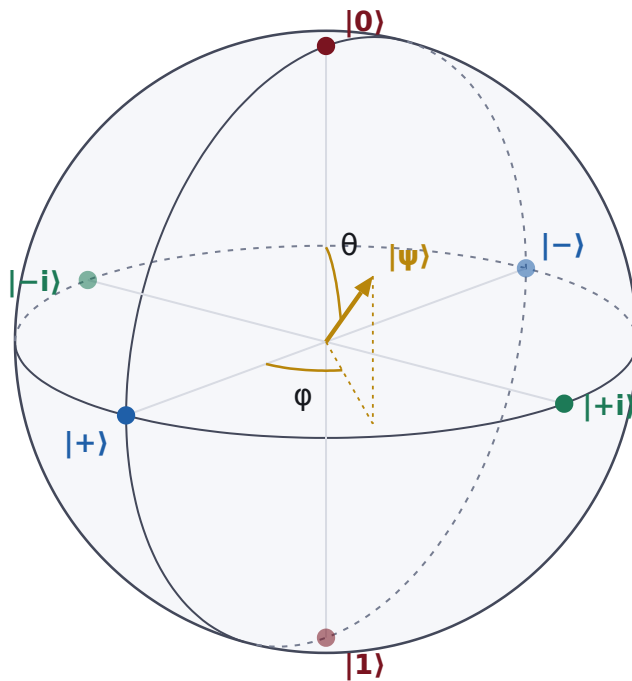
BEYOND THE SYLLABUS · EVERY QUBIT STATE IN TWO ANGLES, AND THE BLOCH SPHERE

A general state $\alpha|0\rangle + \beta|1\rangle$ has two complex numbers, so four real numbers. Normalisation removes one, and the freedom to discard a global phase removes another. Use that freedom to make α real and non-negative. Then every qubit state can be written

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi < 2\pi.$$

Read θ and φ as the polar angle and the azimuth of a point on a sphere of radius 1, the **Bloch sphere** (Figure 3.2). The north pole ($\theta = 0$) is $|0\rangle$ and the south pole ($\theta = \pi$) is $|1\rangle$. The equator ($\theta = \pi/2$) holds the equal superpositions: $|+\rangle$ at $\varphi = 0$, $|+i\rangle$ at $\varphi = \pi/2$, $|-\rangle$ at $\varphi = \pi$ and $| -i\rangle$ at $\varphi = 3\pi/2$.

The half-angles are the reason orthogonal states sit at opposite ends of a diameter: $|0\rangle$ and $|1\rangle$ are at right angles as vectors in $\mathbb{C}^2$ but 180° apart on the sphere. The relative phase is the longitude φ . The global phase has no place on the sphere at all, which is the picture of Postulate 1.



The Bloch sphere

$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$
is the point at polar angle θ
and azimuth φ .

Poles: $|0\rangle$ (top), $|1\rangle$ (bottom)

x-axis: $|+\rangle$ and $|-\rangle$

y-axis: $|+i\rangle$ and $| -i\rangle$

Orthogonal states sit at opposite ends of a diameter, so a 90° angle between kets is 180° on the sphere.

Figure 3.2 The Bloch sphere (beyond the syllabus). Each qubit state is one point on the surface. The six named states sit where the three axes meet the sphere, and the example state $|\psi\rangle$ is drawn at $\theta = 60^\circ$, $\varphi = 50^\circ$. The dashed curves are on the far side.

WORKED EXAMPLE · 6. FINDING THE BLOCH ANGLES (EXTENSION)

$|+i\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$. Match $\cos\frac{\theta}{2} = \frac{1}{\sqrt{2}}$, so $\frac{\theta}{2} = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$. Then $e^{i\varphi}\sin\frac{\pi}{4} = \frac{i}{\sqrt{2}}$ gives $e^{i\varphi} = i$, so $\varphi = \frac{\pi}{2}$. The state is on the equator, a quarter turn from $|+\rangle$.

$|-\rangle$. Again $\theta = \frac{\pi}{2}$, and $e^{i\varphi} = -1$ gives $\varphi = \pi$.

COMMON MISCONCEPTION · A QUBIT IS 0 AND 1 AT THE SAME TIME

You will read this in newspapers and hear it in talks. It is wrong in a way that causes real mistakes. The state $|+\rangle$ is one definite state: a particular unit vector. It is exactly as definite as $|0\rangle$, and in the $\{|+\rangle, |-\rangle\}$ basis it gives the same answer every time you measure it. What is true is that if you measure $|+\rangle$ **in the computational basis**, you get 0 half the time and 1 half the time. “Superposition” is a statement about which basis you wrote the vector in. Also, if a qubit were “both 0 and 1”, then $|+\rangle$ and $|-\rangle$ would both be “both”, and that phrase cannot explain why they are perfectly distinguishable.

Quantum measurement

The Born rule in the computational basis

Postulate 3 is where the numbers of Units 1 and 2 turn into predictions. Start with the simplest case: measure a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the computational basis.

DEFINITION · BORN RULE (COMPUTATIONAL BASIS)

Measuring $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the basis $\{|0\rangle, |1\rangle\}$ gives

outcome 0 with probability $p(0) = |\langle 0|\psi\rangle|^2 = |\alpha|^2$, outcome 1 with probability $p(1) = |\langle 1|\psi\rangle|^2 = |\beta|^2$.

Immediately after the measurement the state is $|0\rangle$ if the outcome was 0, and $|1\rangle$ if it was 1.

The probabilities add to 1 **because the state is normalised**: $p(0) + p(1) = |\alpha|^2 + |\beta|^2 = 1$. This is the reason Postulate 1 insists on unit vectors. A ket of length 2 would predict probabilities adding to 4.

WORKED EXAMPLE · 7. COMPUTATIONAL-BASIS PROBABILITIES

Measure $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$ in the computational basis.

$$p(0) = \left|\frac{3}{5}\right|^2 = \frac{9}{25} = 0.36, \quad p(1) = \left|\frac{4i}{5}\right|^2 = \frac{(-4i)(4i)}{25} = \frac{16}{25} = 0.64.$$

Check: $0.36 + 0.64 = 1$. Notice that the i in the second amplitude changed nothing here. The modulus squared removes the phase of each amplitude separately, so the computational basis cannot see relative phase. Another basis can, as you will see shortly.

PITFALL · THE AMPLITUDE IS NOT THE PROBABILITY

The bracket $\langle b|\psi\rangle$ is an **amplitude**: a complex number that can be negative or imaginary. $|\langle b|\psi\rangle|^2$ is a **probability**: a real number between 0 and 1. For $|\psi\rangle$ above, $\langle 1|\psi\rangle = 0.8i$ and the probability is 0.64. An answer like "the probability is $0.8i$ " or "the probability is -0.5 " means you forgot to take the modulus squared.

Collapse and repeated measurement

The second half of Postulate 3 says the state changes. If you measure $\frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$ and get outcome 1, the qubit is now in the state $|1\rangle$. The amplitudes $\frac{3}{5}$ and $\frac{4i}{5}$ are gone. This is called the **collapse** of the state.

Collapse makes a prediction you can test: measure the same qubit again, in the same basis, straight away. The state is $|1\rangle$, so $p(1) = |\langle 1|1\rangle|^2 = 1$. You get 1 again, with certainty, however many times you repeat. A second measurement in the same basis only confirms the first.

COMMON MISCONCEPTION · MEASUREMENT JUST READS OFF A VALUE THE QUBIT ALREADY HAD

It is natural to think that a qubit in $\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$ secretly “is” 0 or 1 all along, and the measurement reveals which. The mathematics says otherwise. The measurement outcome depends on the basis you choose, and measuring in one basis destroys the information another basis would have given (Worked example 11 shows this). Going further, Bell’s theorem, which uses entangled pairs like the ones later in this unit, shows that no theory in which each particle carries pre-set answers to every possible measurement can reproduce all the predictions of quantum mechanics. Experiments side with quantum mechanics. Treat the outcome as something the measurement *produces*, with probabilities set by the state.

Measuring in the $\{|+\rangle, |-\rangle\}$ basis

Nothing in Postulate 3 says you must measure in the computational basis. Any orthonormal basis will do, and in the laboratory choosing a basis means choosing which question to ask (for a photon, which angle to set the polariser to). The rule is the same: the probability of outcome b_k is $|\langle b_k | \psi \rangle|^2$.

For the $\pm$ basis, with $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

$$\langle + | \psi \rangle = \frac{1}{\sqrt{2}}(\alpha + \beta), \quad \langle - | \psi \rangle = \frac{1}{\sqrt{2}}(\alpha - \beta),$$

so

$$p(+)=\frac{1}{2}|\alpha+\beta|^2, \quad p(-)=\frac{1}{2}|\alpha-\beta|^2.$$

WORKED EXAMPLE · 8. ONE STATE, TWO BASES

Take the real state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$.

Computational basis: $p(0) = 0.36$, $p(1) = 0.64$.

$\pm$ basis: $p(+)=\frac{1}{2}\left(\frac{3}{5}+\frac{4}{5}\right)^2=\frac{1}{2}\cdot\frac{49}{25}=\frac{49}{50}=0.98$ and $p(-)=\frac{1}{2}\left(\frac{3}{5}-\frac{4}{5}\right)^2=\frac{1}{2}\cdot\frac{1}{25}=\frac{1}{50}=0.02$.

The same state is close to a coin toss in one basis and nearly certain in the other (Figure 3.3). Looking back at Figure 3.1, this makes sense: the arrow for $|\psi\rangle$ lies close to the arrow for $|+\rangle$ and almost at right angles to $|-\rangle$.

Now the complex state $|\phi\rangle = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$. It has the same computational-basis probabilities, 0.36 and 0.64. But

$$p(+)=\frac{1}{2}\left|\frac{3}{5}+\frac{4i}{5}\right|^2=\frac{1}{2}\cdot\frac{9+16}{25}=\frac{1}{2}, \quad p(-)=\frac{1}{2}.$$

The states $|\psi\rangle$ and $|\phi\rangle$ differ only by a relative phase i on the $|1\rangle$ amplitude. The computational basis cannot tell them apart; the $\pm$ basis can, giving 0.98 against 0.50.

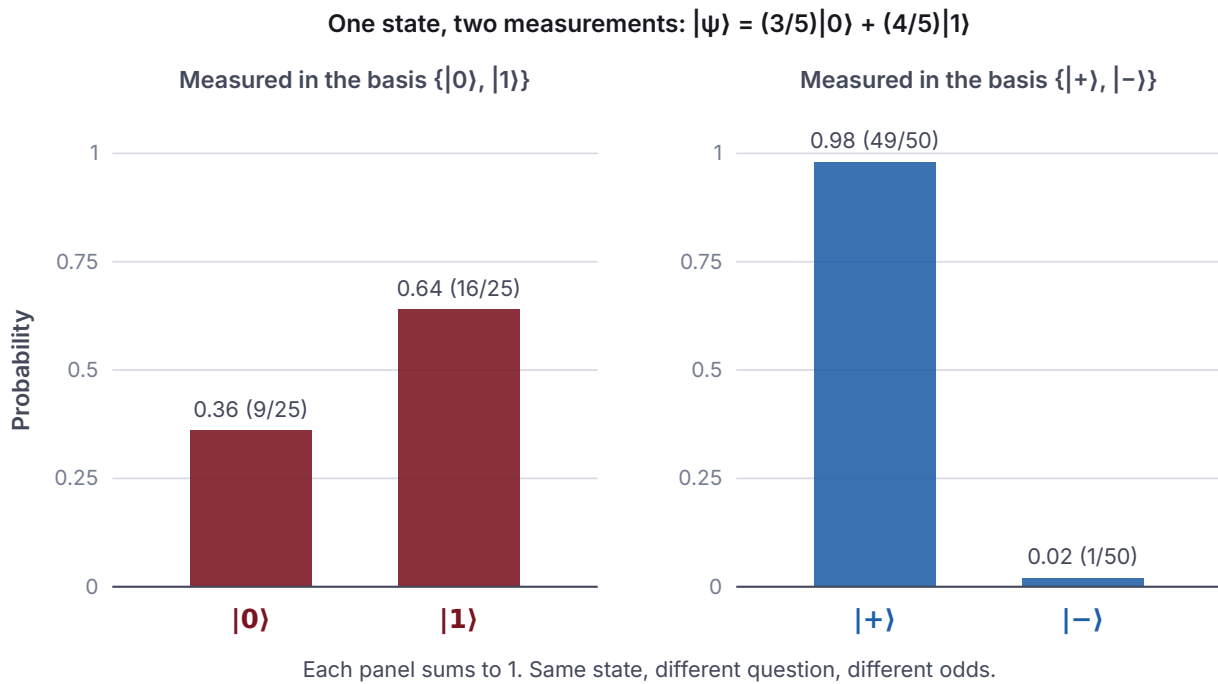


Figure 3.3 The Born rule for $|\psi\rangle = (3/5)|0\rangle + (4/5)|1\rangle$ in two bases. Each panel is a complete probability distribution and adds to 1. The state has not changed between the panels; only the question has.

Measuring in any orthonormal basis

The general statement is the one in the postulates table. Given an orthonormal basis $\{|b_1\rangle, \dots, |b_n\rangle\}$:

$$p(k) = |\langle b_k | \psi \rangle|^2, \quad \text{state afterwards } |b_k\rangle.$$

The probabilities always add to 1. The completeness relation gives $|\psi\rangle = \sum_k \langle b_k | \psi \rangle |b_k\rangle$, and the norm squared of a vector in an orthonormal basis is the sum of its squared components:

$$\sum_k p(k) = \sum_k |\langle b_k | \psi \rangle|^2 = \langle \psi | \psi \rangle = 1.$$

The basis must be orthonormal. If it were not, these probabilities would not add to 1 and the rule would make no sense.

WORKED EXAMPLE · 9. A THIRD BASIS

The bras are $\langle +i| = \frac{1}{\sqrt{2}}(1, -i)$ and $\langle -i| = \frac{1}{\sqrt{2}}(1, i)$. Remember the conjugation.

(a) Measure $|+\rangle$. $\langle +i|+\rangle = \frac{1}{2}(1 - i)$, so $p(+i) = \frac{1}{4}|1 - i|^2 = \frac{1}{4} \cdot 2 = \frac{1}{2}$, and $p(-i) = \frac{1}{2}$.

(b) Measure $|+i\rangle$. $\langle +i|+i\rangle = 1$, so $p(+i) = 1$. This basis gives different statistics for $|+\rangle$ and $|+i\rangle$, which confirms Worked example 5(c): those are different states.

(c) Measure $|\phi\rangle = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$. $\langle +i|\phi\rangle = \frac{1}{\sqrt{2}}(\frac{3}{5} + (-i)\frac{4i}{5}) = \frac{1}{\sqrt{2}} \cdot \frac{7}{5}$, so $p(+i) = \frac{49}{50} = 0.98$ and $p(-i) = 0.02$. The complex state behaves in this basis exactly as the real state $\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$ did in the $\pm$ basis.

Statistics: what “probability” means here

A single measurement gives a single outcome, 0 or 1. The probability $p(0) = 0.36$ cannot be seen in one shot. It is a statement about many repetitions: prepare the same state again and again, measure each copy once, and count. As the number of copies grows, the fraction of 0s gets closer and closer to $p(0)$.

WORKED EXAMPLE · 10. A THOUSAND COPIES

A source prepares $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$ a thousand times, and each copy is measured once in the computational basis.

Expected counts: $1000 \times 0.36 = 360$ zeros and $1000 \times 0.64 = 640$ ones.

What you actually see is close to these numbers but rarely equal to them. Counts such as 366 and 634 are typical; the random scatter for a thousand copies is a couple of dozen either way. A result of 500 and 500 would be far outside that scatter, and you should conclude that the source is not making the state it claims.

What you can infer: $|\alpha|^2 \approx 0.37$ and $|\beta|^2 \approx 0.63$. You learn nothing about the phases of α and β from these counts. To find the relative phase you need a second set of copies measured in a different basis, such as the $\pm$ basis.

NOTE · CHECK IT IN NUMPY: THE BORN RULE AND SAMPLING

```
import numpy as np
psi = np.array([3/5, 4/5])
probs = np.abs(psi)**2           # Born rule: [0.36 0.64]
rng = np.random.default_rng(7)
outcomes = rng.choice([0, 1], size=1000, p=probs)
print(probs, np.bincount(outcomes))  # [0.36 0.64] [366 634]
plus, minus = np.array([1, 1])/np.sqrt(2), np.array([1, -1])/np.sqrt(2)
print(abs(np.vdot(plus, psi))**2, abs(np.vdot(minus, psi))**2) # 0.98 0.02
```

Change the seed, or remove it, and the counts change while staying near 360 and 640. This is how the simulator in Part C of the assignment will “measure” a qubit.

What measurement cannot tell you

Put the last few results together and you get the most important practical fact about qubits: **you cannot read out a state**. A single measurement of one copy gives one bit, 0 or 1, and destroys the amplitudes in the process. From one outcome you cannot recover α and β ; you cannot even tell whether the state was $|0\rangle$ or $\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$, since both can give 0. Estimating the state takes many identically prepared copies, measured in more than one basis.

WORKED EXAMPLE · 11. TELLING $|+\rangle$ FROM $|-\rangle$

A friend hands you one qubit and promises it is either $|+\rangle$ or $|-\rangle$.

Measure in the computational basis. For $|+\rangle$: $p(0) = p(1) = \frac{1}{2}$. For $|-\rangle$: $p(0) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$ and $p(1) = \left|-\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$. The statistics are identical, so the outcome tells you nothing. Worse, the qubit has collapsed to $|0\rangle$ or $|1\rangle$, and the information is gone for good.

Measure in the $\pm$ basis. For $|+\rangle$: $p(+) = |\langle +|+\rangle|^2 = 1$. For $|-\rangle$: $p(+) = |\langle +|-\rangle|^2 = 0$. You get the right answer every time.

The only difference between $|+\rangle$ and $|-\rangle$ is a relative phase. The right choice of basis turns that phase into a certain, visible difference. The wrong choice hides it completely.

Quantum operations

Postulate 2: unitary evolution

Between preparation and measurement, a closed system changes by a unitary matrix:

$$|\psi\rangle \longrightarrow |\psi'\rangle = U|\psi\rangle, \quad U^\dagger U = I.$$

“Closed” means the system is not being measured and is not interacting with anything else. In quantum computing a unitary applied to qubits is called a **quantum gate**, and a sequence of gates followed by measurements is a **quantum circuit**.

Why unitary? Because the output must again be a valid state. If $|\psi\rangle$ is normalised, so is $U|\psi\rangle$:

$$\langle\psi'|\psi'\rangle = (U|\psi\rangle)^\dagger (U|\psi\rangle) = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|I|\psi\rangle = \langle\psi|\psi\rangle = 1.$$

Here $(U|\psi\rangle)^\dagger = \langle\psi|U^\dagger$, the dagger of a product reversing the order, as in Unit 2. A unitary therefore keeps total probability equal to 1, which any physical process has to do. A unitary also has an inverse, $U^{-1} = U^\dagger$, so every gate can be undone. Measurement, by contrast, is not unitary and cannot be undone.

Five gates

GATE	MATRIX	$ 0\rangle \rightarrow$	$ 1\rangle \rightarrow$	$ +\rangle \rightarrow$	$ -\rangle \rightarrow$
X (bit flip)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$ 1\rangle$	$ 0\rangle$	$ +\rangle$	$- -\rangle$
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	$i 1\rangle$	$-i 0\rangle$	$-i -\rangle$	$i +\rangle$
Z (phase flip)	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$ 0\rangle$	$- 1\rangle$	$ -\rangle$	$ +\rangle$
H (Hadamard)	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$
S (phase)	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$ 0\rangle$	$i 1\rangle$	$ +i\rangle$	$ -i\rangle$

The gates X , Y and Z are the **Pauli gates**. Each of the five is unitary; you checked some of them in Unit 2, and H and S are one line each. X , Y , Z and H are also Hermitian, a fact that returns in the section on observables.

Read the table through the lens of Postulate 1. X swaps $|0\rangle$ and $|1\rangle$, the quantum version of a NOT gate. Z leaves $|0\rangle$ alone and puts a minus sign on $|1\rangle$. On a basis state that sign is a global phase and changes nothing, but on a superposition it is a relative phase, and Z turns $|+\rangle$ into $|-\rangle$. Likewise

$X|-\rangle = -|-\rangle$ is the **same physical state** as $|-\rangle$. H swaps the computational basis with the $\pm$ basis, and S adds a quarter-turn relative phase.

WORKED EXAMPLE · 12. FILLING IN TWO TABLE ENTRIES

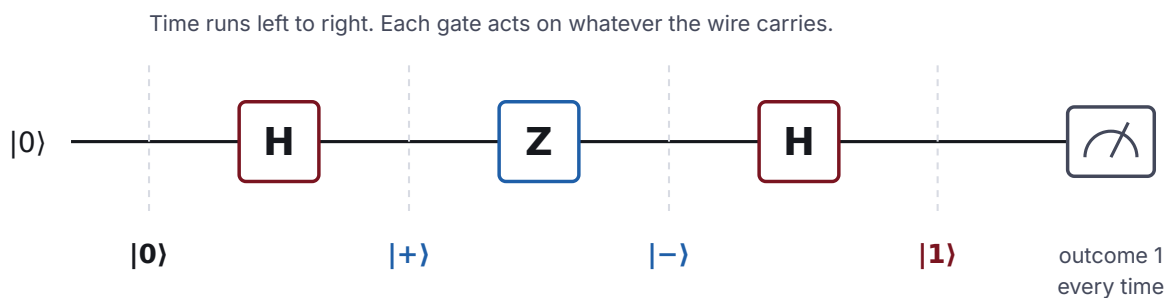
$S|+\rangle$. S multiplies the second amplitude by i : $S \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = |+i\rangle$.

$Y|+\rangle$. Work with basis states and use linearity: $Y|+\rangle = \frac{1}{\sqrt{2}}(Y|0\rangle + Y|1\rangle) = \frac{1}{\sqrt{2}}(i|1\rangle - i|0\rangle) = -i \cdot \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = -i|-\rangle$. Up to the global phase $-i$, Y takes $|+\rangle$ to $|-\rangle$.

Sequences of gates

Apply A first and then B . The state goes $|\psi\rangle \rightarrow A|\psi\rangle \rightarrow B(A|\psi\rangle)$, so the combined operation is the matrix product BA . **The gate applied first is written on the right**, next to the ket. A product of unitaries is unitary, so a whole circuit is one big unitary.

In a circuit diagram (Figure 3.4) time runs left to right: the first gate is on the left. In the matrix formula the first gate is on the right. Both are correct; they are read in opposite directions.



As matrices: $H Z H |0\rangle$ (the rightmost matrix acts first) $= X|0\rangle = |1\rangle$

Figure 3.4 A one-qubit circuit. Starting from $|0\rangle$, the gates H , Z and H are applied in that order and the qubit is then measured. The state between the gates is written under each dashed line. Written as matrices the circuit is $H Z H$, with the first H on the right.

PITFALL · GATE ORDER

The circuit “ H , then Z , then H ” is the matrix $H Z H$ acting as $H Z H |\psi\rangle$. The circuit “ X , then H ” is $H X |\psi\rangle$, **not** $X H |\psi\rangle$. Matrices do not commute in general (Unit 2, session 5), so writing the product in circuit order gives a wrong answer that looks perfectly reasonable. When in doubt, apply the gates one at a time to the ket, as in the next example.

WORKED EXAMPLE · 13. $HZH = X$ AND $H^2 = I$

Multiply out, remembering $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$:

$$ZH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad HZH = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = X.$$

So the circuit of Figure 3.4 is a bit flip in disguise: $HZH|0\rangle = X|0\rangle = |1\rangle$. Following the kets gives the same answer: $|0\rangle \rightarrow |+\rangle \rightarrow |-\rangle \rightarrow |1\rangle$.

Similarly $H^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I$. Applying H twice does nothing. Since H is Hermitian, $H^\dagger = H$, and $H^2 = I$ is the same statement as " H is unitary".

WORKED EXAMPLE · 14. ORDER MATTERS

Start from $|0\rangle$.

X, then H: $HX|0\rangle = H|1\rangle = |-\rangle$.

H, then X: $XH|0\rangle = X|+\rangle = |+\rangle$.

The two orders give orthogonal states, which a $\pm$ -basis measurement separates every time. In the computational basis both give 0 and 1 with probability $\frac{1}{2}$ each, so that measurement would hide the difference.

Interference: amplitudes add, and can cancel

Apply H to $|0\rangle$ and then apply H again. You already know the answer, $H^2 = I$, so you get $|0\rangle$ back. It is worth watching how.

After the first H : $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$. A measurement now would give 0 or 1 with probability $\frac{1}{2}$ each.

Apply the second H to each term, using $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$H|+\rangle = \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) = \frac{1}{2}(|0\rangle + |1\rangle) + \frac{1}{2}(|0\rangle - |1\rangle) = \underbrace{\left(\frac{1}{2} + \frac{1}{2}\right)}_1 |0\rangle + \underbrace{\left(\frac{1}{2} - \frac{1}{2}\right)}_0 |1\rangle = |0\rangle.$$

Both branches contribute an amplitude to $|1\rangle$, one $+\frac{1}{2}$ and one $-\frac{1}{2}$, and they **cancel**. Both contribute $+\frac{1}{2}$ to $|0\rangle$, and those **add**. This adding and cancelling of amplitudes is called **interference**. It is the reason relative phases matter: the sign on the $|1\rangle$ amplitude decided which outcome survived. Start from $|-\rangle$ instead and the signs arrange themselves the other way, giving $H|-\rangle = |1\rangle$.

PITFALL · ADDING PROBABILITIES INSTEAD OF AMPLITUDES

Here is the wrong argument. "After the first H the qubit is 0 or 1 with probability $\frac{1}{2}$ each. Each of those, after the second H , gives 0 or 1 with probability $\frac{1}{2}$. So the final result is 0 or 1 with probability $\frac{1}{2}$ each." That prediction is wrong: the real answer is 0 with certainty.

The error is combining **probabilities** at the middle step. Between measurements you must combine **amplitudes**, which carry signs and phases and can cancel. Probabilities are never negative, so they can only pile up. Take the modulus squared once, at the end, when the measurement happens.

The wrong argument *does* describe a different experiment: measure after the first H , then apply the second H . The measurement collapses the state to $|0\rangle$ or $|1\rangle$, the second H gives $|+\rangle$ or $|-\rangle$, and the final outcome really is 50:50. Measuring in the middle destroys the interference.

NOTE · CHECK IT IN NUMPY: GATES AND CIRCUITS

```
import numpy as np
X = np.array([[0, 1], [1, 0]])
Z = np.array([[1, 0], [0, -1]])
H = np.array([[1, 1], [1, -1]]) / np.sqrt(2)
S = np.array([[1, 0], [0, 1j]])
ket0 = np.array([1, 0])
print(np.allclose(H @ Z @ H, X))           # True
print(H @ Z @ H @ ket0)                   # about [0, 1], that is |1>
print(np.allclose(S.conj().T @ S, np.eye(2))) # S is unitary
print(H @ X @ ket0, X @ H @ ket0)         # |-> and |+>
```

@ is matrix multiplication. The expression `H @ Z @ H @ ket0` is read exactly as the maths: the gate next to the ket acts first. NumPy may print a tiny number such as `-2.2e-17` where you expect 0. That is rounding error from $\frac{1}{\sqrt{2}}$, and it is why the check uses `np.allclose` instead of `==`.

Multiple quantum states

Postulate 4: the tensor product

One qubit lives in $\mathbb{C}^2$. Postulate 4 says two qubits together live in the tensor product

$$\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4.$$

The computational basis of two qubits is the four products of single-qubit basis states. We write $|ab\rangle = |a\rangle \otimes |b\rangle$, where the first (leftmost) symbol always belongs to **qubit 1**:

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

The order is the order of counting in binary, 00, 01, 10, 11. It is exactly the Kronecker product of Unit 2, session 9:

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}.$$

A general two-qubit state is any unit vector in $\mathbb{C}^4$:

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle, \quad |c_{00}|^2 + |c_{01}|^2 + |c_{10}|^2 + |c_{11}|^2 = 1.$$

Measuring both qubits in the computational basis gives the outcome ab with probability $|c_{ab}|^2$. That is Postulate 3 again, in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

PITFALL · WHICH QUBIT IS WHICH

In $|01\rangle$, qubit 1 is in $|0\rangle$ and qubit 2 is in $|1\rangle$. It is the **second** entry of the four-vector. In NumPy, `np.kron(a, b)` puts `a` as qubit 1, so the order of the arguments matters: `np.kron(ket0, ket1)` is $|01\rangle$ while `np.kron(ket1, ket0)` is $|10\rangle$. Some software (Qiskit, for one) numbers qubits the other way round. If a result looks mirrored, check the ordering convention before anything else.

Product states

When each qubit has a state of its own, $|u\rangle$ and $|v\rangle$, the joint state is the **product state** $|u\rangle \otimes |v\rangle$.

WORKED EXAMPLE · 15. BUILDING PRODUCT STATES

(a) $|+\rangle \otimes |0\rangle$. With the Kronecker product, $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, which is $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$. Expanding the kets directly gives the same thing: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|0\rangle)$.

(b) In general,

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

A product of two unit vectors is a unit vector, since the probabilities multiply: $(|a|^2 + |b|^2)(|c|^2 + |d|^2) = 1 \cdot 1$.

Measuring one qubit of two

Often you measure only qubit 1. Postulate 3 then says: add up the probabilities of all basis states that agree with the outcome, and keep only those terms afterwards.

$$p(\text{qubit 1 gives } 0) = |c_{00}|^2 + |c_{01}|^2, \quad \text{state afterwards } \frac{c_{00}|00\rangle + c_{01}|01\rangle}{\sqrt{|c_{00}|^2 + |c_{01}|^2}}.$$

The division by the square root renormalises what is left, so it is again a unit vector. Take this as the working rule; it follows from Postulate 3 with a projector in place of a single basis vector.

WORKED EXAMPLE · 16. MEASURING ONE QUBIT OF A PRODUCT STATE

Let $|\Psi\rangle = (\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle) \otimes |+\rangle$. Expanding, the amplitudes are $c_{00} = c_{01} = \frac{3}{5\sqrt{2}}$ and $c_{10} = c_{11} = \frac{4}{5\sqrt{2}}$.

Then $p(\text{qubit 1 gives } 0) = \frac{9}{50} + \frac{9}{50} = \frac{9}{25} = 0.36$, exactly the probability you would compute from qubit 1 alone. After outcome 0 the state is $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes |+\rangle$. Qubit 2 is still $|+\rangle$, untouched. For a product state, measuring one qubit tells you nothing about the other and does not disturb it.

Gates on one qubit of two

To apply a gate A to qubit 1 and nothing to qubit 2, use the 4×4 matrix $A \otimes I$. To act on qubit 2 only, use $I \otimes A$. On product states the Kronecker product does what you would hope:

$$(A \otimes B)(|u\rangle \otimes |v\rangle) = A|u\rangle \otimes B|v\rangle.$$

WORKED EXAMPLE · 17. $X \otimes I$ AND $I \otimes X$

$$X \otimes I = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad I \otimes X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Check: $(X \otimes I)|00\rangle = X|0\rangle \otimes |0\rangle = |10\rangle$, and the first column of $X \otimes I$ is indeed $|10\rangle$. And $(I \otimes X)|00\rangle = |01\rangle$. If your matrix for “flip qubit 1” sends $|00\rangle$ to $|01\rangle$, you have the Kronecker factors in the wrong order.

CNOT: a gate on two qubits at once

The controlled-NOT gate flips qubit 2 **if and only if** qubit 1 is $|1\rangle$. Qubit 1 is the **control**, qubit 2 the **target**.

INPUT	OUTPUT
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

The columns of a matrix are the images of the basis vectors, so

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

It is unitary (it is a permutation matrix, and it is its own inverse). CNOT cannot be written as $A \otimes B$ for any single-qubit gates A and B , and that is what makes it useful: it is how two qubits come to depend on each other.

Creating a Bell state

Start with both qubits in $|0\rangle$, apply H to qubit 1, then apply CNOT (Figure 3.5).

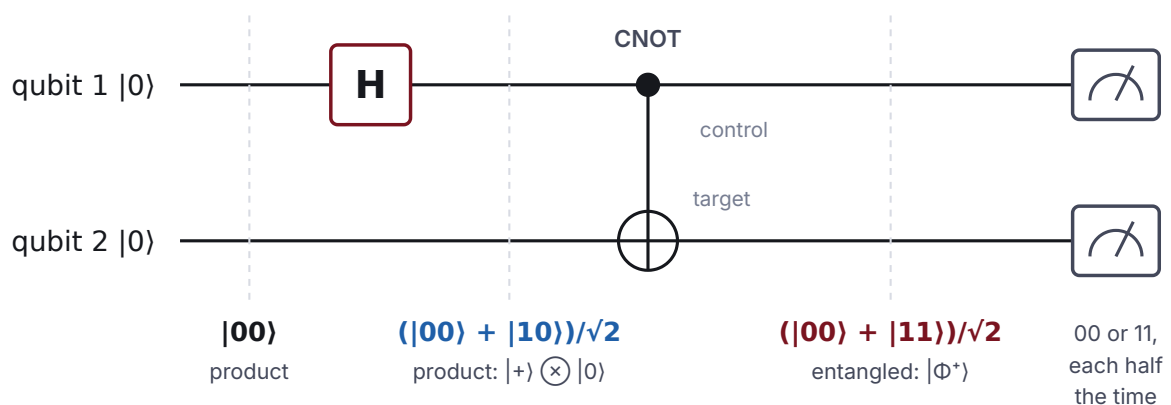


Figure 3.5 The Bell-state circuit. The filled dot marks the control qubit and $\oplus$ the target. The state after each stage is written underneath. As matrices the circuit is $\text{CNOT} \cdot (H \otimes I)$ acting on $|00\rangle$.

WORKED EXAMPLE · 18. $|00\rangle \rightarrow |\Phi^+\rangle$, STEP BY STEP

Start. $|00\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}^T$.

Step 1, H on qubit 1. $(H \otimes I)|00\rangle = H|0\rangle \otimes |0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \end{pmatrix}^T$. This is still a product state.

Step 2, CNOT. CNOT leaves $|00\rangle$ alone and sends $|10\rangle$ to $|11\rangle$. By linearity,

$$\text{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle.$$

As a matrix product, CNOT swaps the third and fourth entries: $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \end{pmatrix}^T \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix}^T$.

Entanglement

DEFINITION · ENTANGLED STATE

A two-qubit state is **entangled** if it **cannot** be written as a product $|u\rangle \otimes |v\rangle$ of two single-qubit states. A state that can be written that way is a product state.

This is the one place in the unit where the definition is a “cannot”, and there is no way round it. So to show a state is entangled you must prove that no choice of $|u\rangle$ and $|v\rangle$ works.

WORKED EXAMPLE · 19. $|\Phi^+\rangle$ IS ENTANGLED

Suppose $|\Phi^+\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$ for some complex a, b, c, d . By Worked example 15(b), matching the four amplitudes of $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ requires

$$ac = \frac{1}{\sqrt{2}}, \quad ad = 0, \quad bc = 0, \quad bd = \frac{1}{\sqrt{2}}.$$

From $ad = 0$, either $a = 0$ or $d = 0$. If $a = 0$ then $ac = 0$, which contradicts $ac = \frac{1}{\sqrt{2}}$. If $d = 0$ then $bd = 0$, which contradicts $bd = \frac{1}{\sqrt{2}}$. Both cases fail, so no such a, b, c, d exist and $|\Phi^+\rangle$ is not a product state. It is entangled.

A quicker test works for any two-qubit state: it is a product exactly when $c_{00}c_{11} = c_{01}c_{10}$. For $|\Phi^+\rangle$ the left side is $\frac{1}{2}$ and the right side is 0.

What entanglement looks like in measurements

Measure both qubits of $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ in the computational basis. The Born rule gives

$$p(00) = \frac{1}{2}, \quad p(01) = 0, \quad p(10) = 0, \quad p(11) = \frac{1}{2}.$$

Each qubit on its own is a fair coin: qubit 1 gives 0 half the time. But the two coins always agree. Measure qubit 1 first and get 0: by the rule above, the state becomes $|00\rangle$, so qubit 2 will give 0 with certainty. Get 1 and the state becomes $|11\rangle$. The outcome of qubit 1 fixes the outcome of qubit 2. Compare Worked example 16, where measuring qubit 1 of a product state left qubit 2 alone.

COMMON MISCONCEPTION · ENTANGLED QUBITS SEND SIGNALS TO EACH OTHER

The correlation is perfect, but nobody can use it to send a message. Whoever holds qubit 1 gets 0 or 1 at random, with probability $\frac{1}{2}$ each, whatever anyone does to qubit 2. Only when the two lists of outcomes are brought together and compared does the agreement show up, and bringing them together is ordinary communication.

The circuit of Figure 3.5 makes a different entangled state from each of the four computational basis inputs. These are the four **Bell states**:

INPUT	OUTPUT OF CNOT · $(H \otimes I)$	NAME
$ 00\rangle$	$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	$ \Phi^+\rangle$
$ 10\rangle$	$\frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$	$ \Phi^-\rangle$
$ 01\rangle$	$\frac{1}{\sqrt{2}}(01\rangle + 10\rangle)$	$ \Psi^+\rangle$
$ 11\rangle$	$\frac{1}{\sqrt{2}}(01\rangle - 10\rangle)$	$ \Psi^-\rangle$

In $|\Phi^\pm\rangle$ the two qubits always agree in the computational basis; in $|\Psi^\pm\rangle$ they always disagree. The four are orthonormal, so they form a basis of $\mathbb{C}^4$, the **Bell basis**.

NOTE · CHECK IT IN NUMPY: TWO QUBITS

```
import numpy as np
ket0 = np.array([1, 0])
H, I = np.array([[1, 1], [1, -1]]) / np.sqrt(2), np.eye(2)
CNOT = np.array([[1, 0, 0, 0], [0, 1, 0, 0],
                 [0, 0, 0, 1], [0, 0, 1, 0]])
ket00 = np.kron(ket0, ket0) # [1 0 0 0]
bell = CNOT @ np.kron(H, I) @ ket00
print(bell) # [0.7071 0. 0. 0.7071]
print(np.abs(bell)**2) # p(00), p(01), p(10), p(11)
```

Many qubits, and why simulation gets hard

Postulate 4 repeats. Three qubits live in $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^8$, and n qubits need a vector of 2^n complex amplitudes. Stored as NumPy `complex128` values (16 bytes each), 30 qubits need $2^{30} \times 16$ bytes, about 17 GB, more memory than most laptops have; fifty qubits need about 18 million GB. A general entangled state cannot be broken into single-qubit pieces, so a classical computer has to carry all 2^n amplitudes. That is why quantum computers are interesting and why they are hard to simulate. The simulator you build in Part C of the Qubit Workbook does exactly this bookkeeping for a few qubits.

Observables

A physical quantity is a Hermitian matrix

So far, a measurement has meant “measure in this orthonormal basis and report which basis vector you got”. Physicists usually describe a measurement by the **quantity** it measures instead: an energy, a spin component, a polarisation. The last piece of the framework says how.

DEFINITION · OBSERVABLE

A physical quantity that can be measured is represented by a **Hermitian** matrix A , called an **observable**. Then:

1. The possible outcomes of measuring A are the **eigenvalues** λ_k of A .
2. Outcome λ_k occurs with probability $p(\lambda_k) = |\langle a_k | \psi \rangle|^2$, where $|a_k\rangle$ is the unit eigenvector for λ_k .
3. Immediately afterwards the state is $|a_k\rangle$.

Compare this with Postulate 3. It is the same rule with the eigenvectors of A as the measurement basis. **Measuring the observable A means measuring in the eigenbasis of A and reporting the eigenvalue as the result.** The Hermitian condition is what makes this work, and you met both reasons in Unit 2:

- The eigenvalues of a Hermitian matrix are **real**, as a measured quantity must be.
- Eigenvectors of a Hermitian matrix for different eigenvalues are **orthogonal**, so once normalised they form an orthonormal basis, which Postulate 3 needs.

Every single-qubit observable in this unit has two different eigenvalues. When an eigenvalue repeats, as it does for the two-qubit observable in Worked example 23, the collapse rule needs a projector in place of a single eigenvector; that refinement belongs to a later course. The expectation-value formula below holds either way.

Z and X as observables

The matrix $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is Hermitian. It is diagonal, so its eigenvectors are the computational basis:

$$Z|0\rangle = +1|0\rangle, \quad Z|1\rangle = -1|1\rangle.$$

Measuring Z is measuring in the computational basis, with the outcome labelled $+1$ for $|0\rangle$ and -1 for $|1\rangle$ in place of 0 and 1. For a spin- $\frac{1}{2}$ particle, Z (times a constant) is the spin component along the z axis, which is where the name comes from.

The matrix $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is also Hermitian. From the gate table, $X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$, so

$$X|+\rangle = +1|+\rangle, \quad X|-\rangle = -1|-\rangle.$$

Measuring X is measuring in the $\pm$ basis, with outcomes $+1$ and -1 . In the same way Y has eigenvalues ± 1 with eigenvectors $|+i\rangle$ and $|-i\rangle$ (self-assessment question 14 asks you to check it).

PITFALL · THE GATE AND THE OBSERVABLE ARE THE SAME MATRIX, USED IN TWO WAYS

Applying the gate Z to a state means multiplying: $|\psi\rangle \rightarrow Z|\psi\rangle$. Measuring the observable Z means using its eigenvectors as a measurement basis. The second does **not** involve computing $Z|\psi\rangle$ and calling that the new state. After measuring Z on $|+\rangle$ the state is $|0\rangle$ or $|1\rangle$, while applying the gate Z to $|+\rangle$ gives $|-\rangle$.

Expectation value

Measure A on many copies of $|\psi\rangle$ and average the outcomes. The average is the **expectation value**, the probability-weighted mean of the eigenvalues:

$$\langle A \rangle = \sum_k \lambda_k p(\lambda_k).$$

Quantum mechanics has a neat formula for it that needs no eigenvectors at all:

$$\langle A \rangle = \langle \psi | A | \psi \rangle.$$

The “sandwich” on the right is a bra, a matrix and a ket multiplied in order, which gives one number. For a Hermitian A that number is real.

WORKED EXAMPLE · 20. THE SANDWICH EQUALS THE AVERAGE, FOR Z

Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Then $Z|\psi\rangle = \alpha|0\rangle - \beta|1\rangle$, and

$$\langle \psi | Z | \psi \rangle = (\alpha^* \quad \beta^*) \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = |\alpha|^2 - |\beta|^2.$$

The outcomes of Z are $+1$ with probability $|\alpha|^2$ and -1 with probability $|\beta|^2$, so the average is $(+1)|\alpha|^2 + (-1)|\beta|^2 = |\alpha|^2 - |\beta|^2$. The two agree for every state.

For $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$: $\langle Z \rangle = 0.36 - 0.64 = -0.28$.

WORKED EXAMPLE · 21. THE SAME CHECK FOR X , WITH NUMBERS

Keep $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$.

Sandwich. $X|\psi\rangle = \frac{4}{5}|0\rangle + \frac{3}{5}|1\rangle$, so $\langle \psi | X | \psi \rangle = \frac{3}{5} \cdot \frac{4}{5} + \frac{4}{5} \cdot \frac{3}{5} = \frac{24}{25} = 0.96$.

Average of outcomes. The eigenbasis of X is $\{|+\rangle, |-\rangle\}$. From Worked example 8, $p(+1) = 0.98$ and $p(-1) = 0.02$, so the average is $(+1)(0.98) + (-1)(0.02) = 0.96$.

Same number, found two ways. The general proof writes $A = \sum_k \lambda_k |a_k\rangle\langle a_k|$ (true for any Hermitian matrix with an orthonormal eigenbasis) and sandwiches: $\langle \psi | A | \psi \rangle = \sum_k \lambda_k \langle \psi | a_k \rangle \langle a_k | \psi \rangle = \sum_k \lambda_k |\langle a_k | \psi \rangle|^2$.

WORKED EXAMPLE · 22. MEASURING Y ON $|0\rangle$

The observable Y has eigenvalues $+1$ (eigenvector $|+i\rangle$) and -1 (eigenvector $|-i\rangle$). On $|0\rangle$, $\langle +i|0\rangle = \frac{1}{\sqrt{2}}$ and $\langle -i|0\rangle = \frac{1}{\sqrt{2}}$, so each outcome has probability $\frac{1}{2}$ and the average is 0. The sandwich agrees: $Y|0\rangle = i|1\rangle$, so $\langle 0|Y|0\rangle = i\langle 0|1\rangle = 0$.

PITFALL · THE EXPECTATION VALUE NEED NOT BE A POSSIBLE OUTCOME

For $|+\rangle$, $\langle Z\rangle = \frac{1}{2} - \frac{1}{2} = 0$, but a single measurement of Z never gives 0. It gives $+1$ or -1 , each half the time, and 0 is their average. In the same way the average score of a fair die is 3.5, which no throw ever shows. Only eigenvalues can appear as outcomes.

NOTE · CHECK IT IN NUMPY: EXPECTATION VALUES

```
import numpy as np
X = np.array([[0, 1], [1, 0]])
psi = np.array([3/5, 4/5])
print(np.vdot(psi, X @ psi))           # 0.96, the sandwich
vals, vecs = np.linalg.eigh(X)         # eigenvalues [-1, 1], eigenvectors as columns
probs = np.abs(vecs.conj().T @ psi)**2 # Born rule in the eigenbasis
print(vals, probs, np.sum(vals * probs)) # [-1 1] [0.02 0.98] 0.96
```

`np.linalg.eigh` is the eigenvalue routine for Hermitian matrices. It returns the eigenvalues in increasing order and the eigenvectors as the columns of `vecs`, so `vecs.conj().T @ psi` computes every $\langle a_k|\psi\rangle$ at once.

BEYOND THE SYLLABUS · SPREAD OF THE OUTCOMES AND A FIRST UNCERTAINTY RELATION

The expectation value is the mean of the outcomes. Their spread is measured by the **variance**, $(\Delta A)^2 = \langle A^2\rangle - \langle A\rangle^2$. For the Pauli matrices $A^2 = I$, so $\langle A^2\rangle = 1$ and $(\Delta A)^2 = 1 - \langle A\rangle^2$.

For $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$, $(\Delta X)^2 = 1 - 0.96^2 = 0.0784$ (small: the X outcome is nearly certain) and $(\Delta Z)^2 = 1 - 0.28^2 = 0.9216$ (large: the Z outcome is nearly a coin toss). They add to 1. For any qubit state one can show $(\Delta X)^2 + (\Delta Z)^2 \geq 1$, so no state gives a certain answer to both questions at once. This is a simple example of an **uncertainty relation**: a limit on how sharply two observables whose matrices do not commute can be known together.

Putting it together

One problem, all four postulates.

WORKED EXAMPLE · 23. FROM SOURCE TO STATISTICS

A source emits qubit 1 in the state $\frac{1}{\sqrt{2}}(i|0\rangle - i|1\rangle)$ and, independently, qubit 2 in $|0\rangle$. A CNOT is applied with qubit 1 as control, and then both qubits are measured in the computational basis.

Postulate 1: the state of qubit 1. It is normalised: $\frac{1}{2}|i|^2 + \frac{1}{2}|-i|^2 = 1$. It equals $i \cdot \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = i|-\rangle$, which is $|-\rangle$ with a global phase i . Physically, qubit 1 is in $|-\rangle$, and we may drop the i .

Postulate 4: the joint state. The qubits were prepared independently, so the joint state is the product $|-\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$, the vector $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix}^T$.

Postulate 2: the CNOT. It leaves $|00\rangle$ alone and sends $|10\rangle$ to $|11\rangle$:

$$\text{CNOT} \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle.$$

Here $c_{00}c_{11} = -\frac{1}{2}$ and $c_{01}c_{10} = 0$, so the state is entangled.

Postulate 3: the measurement. The Born rule gives $p(00) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$, $p(11) = |-\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$ and $p(01) = p(10) = 0$. The minus sign has no effect on these probabilities. If qubit 1 is read first and gives 1, the state collapses to $|11\rangle$ and qubit 2 gives 1 with certainty.

Observables: do the qubits agree? The two-qubit observable $Z \otimes Z = \text{diag}(1, -1, -1, 1)$ is Hermitian. Its value is $+1$ on $|00\rangle$ and $|11\rangle$ (the qubits agree) and -1 on $|01\rangle$ and $|10\rangle$ (they disagree). Then

$$\langle \Phi^- | Z \otimes Z | \Phi^- \rangle = \frac{1}{2}(+1) + \frac{1}{2}(+1) = 1,$$

the largest possible value: the qubits agree on every run. By contrast $\langle Z \otimes I \rangle = \frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$, so qubit 1 on its own is a fair coin.

The postulates in one table

POSTULATE	FORMULA	WHERE YOU USE IT
1 State	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$, $ \alpha ^2 + \beta ^2 = 1$; $e^{i\gamma} \psi\rangle \equiv \psi\rangle$	Normalising; spotting global against relative phase
2 Evolution	$ \psi\rangle \rightarrow U \psi\rangle$, $U^\dagger U = I$; gates in sequence: $\dots U_3 U_2 U_1 \psi\rangle$	Gates and circuits
3 Measurement	$p(k) = \langle b_k \psi \rangle ^2$; afterwards $ b_k\rangle$	Probabilities, collapse, choice of basis
4 Composite systems	$ ab\rangle = a\rangle \otimes b\rangle$; n qubits: $\mathbb{C}^{2^n}$	Two-qubit states, $A \otimes I$, CNOT, entanglement
Observables	$A = A^\dagger$; outcomes λ_k ; $\langle A \rangle = \langle \psi A \psi \rangle = \sum_k \lambda_k p(\lambda_k)$	Measuring a physical quantity, averages

Self-assessment

Try each question on paper first. Then check with NumPy where it helps, then read the solution.

1. In one or two sentences, say what a postulate is and why the Born rule cannot be derived from the mathematics of Unit 2.
2. For $|\psi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2i|1\rangle)$, write the bra $\langle\psi|$ as a row vector and compute $\langle\psi|\psi\rangle$.
3. Compute $\langle-|+i\rangle$ and $\langle+i|-\rangle$, and check that one is the conjugate of the other.
4. Write $|1\rangle\langle 0|$ and $|-\rangle\langle -|$ as matrices. What does $|1\rangle\langle 0|$ do to $|0\rangle$ and to $|1\rangle$?
5. Normalise $2|0\rangle - (1+i)|1\rangle$ and give its computational-basis measurement probabilities.
6. Which pairs describe the same physical state? (a) $|1\rangle$ and $-|1\rangle$; (b) $|+\rangle$ and $|-\rangle$; (c) $\frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$ and $\frac{1}{\sqrt{2}}(i|0\rangle + |1\rangle)$; (d) $|+i\rangle$ and $| -i\rangle$.
7. A qubit is in $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}e^{i\pi/4}|1\rangle$. Give the computational-basis probabilities. The outcome is 1; the qubit is measured again in the same basis. What are the probabilities now?
8. Measure $|0\rangle$ in the $\pm$ basis. Whatever the result, the qubit is then measured in the computational basis. What is the probability of getting 0 at the second measurement?
9. Measure $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ in the $\pm$ basis. Give exact probabilities and decimals.
10. A student computes $p(+)$ for $\frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$ and gets 0.98. Find the mistake and give the right answer.
11. A thousand copies of $\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$ are measured in the computational basis. What counts do you expect? Would 262 zeros surprise you? Would 500?
12. You are given one qubit that is either $|0\rangle$ or $|+\rangle$. Can a computational-basis measurement tell you which, with certainty? What can it tell you?
13. Show that S is unitary. Compute S^2 and $S^\dagger$.
14. Compute $Y|+i\rangle$ and $Y|-i\rangle$. What are the eigenvalues and eigenvectors of Y ?
15. Compute XZ and ZX and show $XZ = -ZX$. For a qubit state $|\psi\rangle$, are $XZ|\psi\rangle$ and $ZX|\psi\rangle$ physically different states?
16. The circuit is H, then S, then H, starting from $|0\rangle$. Write it as a matrix product, find the final state, and give the computational-basis probabilities.
17. Write $|-\rangle \otimes |1\rangle$ as a combination of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ and as a four-vector.
18. Is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ a product state? Is $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$? Prove your answers.
19. Compute $(I \otimes X)|\Phi^+\rangle$. Which Bell state is it?

20. Apply CNOT to $|1\rangle \otimes |+\rangle$ and to $|-\rangle \otimes |-\rangle$. What surprises you about the second?
21. Both qubits of $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ are measured in the computational basis. List the outcome probabilities. If qubit 1 gives 0, what will qubit 2 give?
22. The observable $M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ is measured on $|0\rangle$. Find the possible outcomes, their probabilities, and $\langle M \rangle$ both ways.
23. For $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$, find $\langle Z \rangle$ and $\langle X \rangle$ as functions of φ .
24. A classmate says “for $|+\rangle$ we found $\langle Z \rangle = 0$, so measuring Z on $|+\rangle$ gives 0”. Explain what is wrong.

1. A postulate is a rule accepted without derivation because its predictions agree with experiment. Unit 2 tells you how to compute $|\langle b|\psi\rangle|^2$, but nothing in linear algebra says that this number is the probability of a laboratory outcome. That link between the mathematics and the world is exactly what the postulate supplies.
2. $\langle\psi| = \frac{1}{\sqrt{5}}(1 \quad -2i)$. Then $\langle\psi|\psi\rangle = \frac{1}{5}(1 \cdot 1 + (-2i)(2i)) = \frac{1}{5}(1 + 4) = 1$.
3. $\langle -| = \frac{1}{\sqrt{2}}(1, -1)$, so $\langle -|+i\rangle = \frac{1}{2}(1 - i)$. $\langle +i| = \frac{1}{\sqrt{2}}(1, -i)$, so $\langle +i|-\rangle = \frac{1}{2}(1 + (-i)(-1)) = \frac{1}{2}(1 + i)$. These are complex conjugates, as $\langle\phi|\psi\rangle^* = \langle\psi|\phi\rangle$ requires.
4. $|1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $|-\rangle\langle -| = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. $|1\rangle\langle 0|0\rangle = |1\rangle$ and $|1\rangle\langle 0|1\rangle = 0 \cdot |1\rangle$, the zero vector. It is not unitary, so it is not a gate.
5. Norm squared: $|2|^2 + |-(1+i)|^2 = 4 + 2 = 6$. The state is $\frac{1}{\sqrt{6}}(2|0\rangle - (1+i)|1\rangle)$, with $p(0) = \frac{4}{6} = \frac{2}{3}$ and $p(1) = \frac{2}{6} = \frac{1}{3}$.
6. (a) Same: global phase -1 . (b) Different: relative phase -1 ; they are orthogonal. (c) Same: $i \cdot \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) = \frac{1}{\sqrt{2}}(i|0\rangle + |1\rangle)$, so the second is the first times the global phase i . (d) Different: they are orthogonal.
7. $p(0) = \frac{1}{3}$ and $p(1) = \left|\sqrt{\frac{2}{3}}e^{i\pi/4}\right|^2 = \frac{2}{3}$; the phase $e^{i\pi/4}$ has modulus 1. After outcome 1 the state is $|1\rangle$, so the second measurement gives 1 with probability 1 and 0 with probability 0.
8. The first measurement gives $|+\rangle$ or $|-\rangle$, each with probability $\frac{1}{2}$. Either way the qubit is now $|+\rangle$ or $|-\rangle$, and each gives 0 with probability $\frac{1}{2}$ in the computational basis. So $p(0) = \frac{1}{2}$. The first measurement wiped out the certainty that $|0\rangle$ would have given.
9. $p(+)=\frac{1}{2}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}\right)^2=\frac{1}{2}\cdot\frac{4+2\sqrt{3}}{4}=\frac{2+\sqrt{3}}{4}\approx 0.933$, and $p(-)=\frac{1}{2}\left(\frac{\sqrt{3}}{2}-\frac{1}{2}\right)^2=\frac{2-\sqrt{3}}{4}\approx 0.067$. They add to 1.
10. The student dropped the i and computed $\frac{1}{2}\left(\frac{3}{5}+\frac{4}{5}\right)^2$. Correctly, $\langle +|\psi\rangle=\frac{1}{\sqrt{2}}\left(\frac{3}{5}+\frac{4i}{5}\right)$ and $p(+)=\frac{1}{2}\cdot\left(\frac{9}{25}+\frac{16}{25}\right)=\frac{1}{2}$. The modulus of a sum of complex numbers is not the sum of their moduli.
11. $p(0)=\frac{1}{4}$, so expect 250 zeros and 750 ones. 262 is well within the normal scatter of a dozen or two and is no surprise. 500 is about 250 away from the expected count, far beyond any reasonable scatter; the source is not making the state it claims.
12. No. If the outcome is 1, the qubit must have been $|+\rangle$, because $|0\rangle$ never gives 1. If the outcome is 0, it could have been either: $|0\rangle$ gives 0 always and $|+\rangle$ gives 0 half the time. The two states are not orthogonal ($\langle 0|+\rangle=\frac{1}{\sqrt{2}}$), and non-orthogonal states can never be told apart with certainty from one copy.

13. $S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$ (transpose, then conjugate). $S^\dagger S = \begin{pmatrix} 1 & 0 \\ 0 & (-i)(i) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$, so S is unitary. $S^2 = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$.

14. $Y|+i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \cdot i \\ i \cdot 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = |+i\rangle$. $Y|-i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} (-i)(-i) \\ i \cdot 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ i \end{pmatrix} = -|-i\rangle$. So Y has eigenvalue $+1$ with eigenvector $|+i\rangle$ and eigenvalue -1 with eigenvector $|-i\rangle$.

15. $XZ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so $XZ = -ZX$. Then $XZ|\psi\rangle = -ZX|\psi\rangle$: the two results differ only by the global phase -1 , so they are the **same** physical state. The matrices do not commute, yet here the order has no physical effect. (Worked example 14 shows a pair where it does.)

16. The matrix product is $HS|0\rangle$, first H on the right. Step by step: $H|0\rangle = |+\rangle$, $S|+\rangle = |+i\rangle$, and $H|+i\rangle = \frac{1}{\sqrt{2}}(|+\rangle + i|-\rangle) = \frac{1}{2}((1+i)|0\rangle + (1-i)|1\rangle)$. Probabilities: $p(0) = \frac{1}{4}|1+i|^2 = \frac{2}{4} = \frac{1}{2}$ and $p(1) = \frac{1}{2}$.

17. $|-\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle)$, the vector $\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & -1 \end{pmatrix}^T$.

18. The first is a product: $|+\rangle \otimes |+\rangle = \frac{1}{2}(1, 1, 1, 1)^T$. Check with the test: $c_{00}c_{11} = \frac{1}{4} = c_{01}c_{10}$. The second is entangled. If it were $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$ we would need $ac = 0$, $ad = \frac{1}{\sqrt{2}}$, $bc = \frac{1}{\sqrt{2}}$, $bd = 0$. From $ac = 0$, $a = 0$ or $c = 0$; the first contradicts $ad \neq 0$ and the second contradicts $bc \neq 0$.

19. $I \otimes X$ flips qubit 2: $|00\rangle \rightarrow |01\rangle$ and $|11\rangle \rightarrow |10\rangle$. So $(I \otimes X)|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = |\Psi^+\rangle$.

20. $|1\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$. CNOT swaps the two terms, which leaves the sum unchanged: the output is $|1\rangle \otimes |+\rangle$, because $X|+\rangle = |+\rangle$. For the second, $|-\rangle \otimes |-\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$. CNOT swaps the $|10\rangle$ and $|11\rangle$ amplitudes, giving $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) = |+\rangle \otimes |-\rangle$. The surprise: the **control** qubit changed, from $|-\rangle$ to $|+\rangle$, while the target did not. "Control" and "target" describe the computational basis only.

21. $p(01) = p(10) = \frac{1}{2}$ and $p(00) = p(11) = 0$. If qubit 1 gives 0, the state collapses to $|01\rangle$ and qubit 2 gives 1 with certainty. The qubits always disagree.

22. M is real and symmetric, so Hermitian. $M|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3|+\rangle$ and $M|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -|-\rangle$. The outcomes are 3 and -1 , with probabilities $|\langle +|0\rangle|^2 = \frac{1}{2}$ and $|\langle -|0\rangle|^2 = \frac{1}{2}$. Average: $3 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 1$. Sandwich: $\langle 0|M|0\rangle$ is the top-left entry, 1. They agree.

23. $\langle Z \rangle = |\alpha|^2 - |\beta|^2 = \frac{1}{2} - \frac{1}{2} = 0$ for every φ . $X|\psi\rangle = \frac{1}{\sqrt{2}}(e^{i\varphi}|0\rangle + |1\rangle)$, so $\langle X \rangle = \frac{1}{2}(e^{i\varphi} + e^{-i\varphi}) = \cos \varphi$ by Euler's formula (Unit 1). The relative phase is invisible to Z and fully visible to X : $\langle X \rangle = 1$ for $|+\rangle$, -1 for $|-\rangle$, and 0 for $|\pm i\rangle$.

24. $\langle Z \rangle$ is the average over many measurements. Each single measurement of Z gives an eigenvalue, $+1$ or -1 , here each with probability $\frac{1}{2}$. The average of many such results is close to 0, but 0 itself never occurs, because 0 is not an eigenvalue of Z .

Where to go next

This unit ends the syllabus. You now have the full rulebook: states are unit vectors, gates are unitaries, measurement follows the Born rule, joint systems use the tensor product, and observables are Hermitian matrices.

Two places put the rulebook to work. The companion course **25AID433 Elements of Quantum Computing** builds circuits and algorithms from exactly these gates and measurements, so every calculation there is a calculation from this booklet. And **Part C of the Qubit Workbook** asks you to write a qubit simulator in NumPy: represent kets as arrays, apply gates as matrices, combine qubits with `np.kron`, and measure by sampling from the Born rule. Every “Check it in NumPy” box in this booklet is a piece of that simulator. Collect them as you go, and test each piece against a hand calculation from the worked examples before you trust it.